

Titration Calculations Worksheet

1. In a titration 12.5 mL of 0.200 M NaOH is required to neutralize 10.0 mL of H_2SO_4 . Calculate the concentration of the acid. (0.125 M)
2. In a titration 22.5 mL of 0.100 M HCl is required to neutralize 20.0 mL of $\text{Ba}(\text{OH})_2$. Calculate the concentration of the base. (0.0563 M)
3. A burette filled with 1.52 M nitric acid solution reads 33.10 mL initially. After titrating a 25.00 mL sample of barium hydroxide the endpoint was reached and the burette showed 46.30 mL. What is the barium hydroxide concentration? (0.402 M)
4. A burette filled with 2.557 M sodium hydroxide solution reads 15.62 mL initially. After titrating a 25.00 mL sample of phosphoric acid the endpoint was reached and the burette now showed 39.22 mL. What is the [phosphoric acid]? (0.8046 M)
5. The following data was obtained during the titration of 1.0097 M sodium hydroxide with a 25.00 mL aliquot of hydrofluoric acid:

	Trial 1	Trial 2	Trial 3
Burette Final Reading	34.56 mL	39.42 mL	44.20 mL
Burette Initial Reading	14.94 mL	19.86 mL	24.66 mL
Vol. of NaOH Added			

Use the above information to determine the concentration of the acid. (0.7896 M)

6. The following data was obtained during the titration of 0.0998 M sodium hydroxide with a 10.00 mL aliquot of sulphuric acid:

	Trial 1	Trial 2	Trial 3
Burette Final Reading	26.05 mL	48.52 mL	33.78 mL
Burette Initial Reading	2.46 mL	34.94 mL	20.22 mL

Vol. of NaOH added

Use the above information to determine the concentration of the acid. (0.0677 M)

7. The following data was obtained during the titration of 2.0554 M hydrochloric acid with a 25.00 mL aliquot of barium hydroxide:

	Trial 1	Trial 2	Trial 3
Burette Final Reading	22.92 mL	25.32 mL	41.30 mL
Burette Initial Reading	0.06 mL	2.58 mL	18.54 mL
Volume of Acid Added			

Use the above information to determine the concentration of the barium hydroxide. (0.9352 M)

8. What volume of 2.549 M NaOH is needed to fully titrate 50.0 mL of 1.285 M HCl solution? (25.2 mL)

9. What volume of 1.146 M KOH is needed to fully titrate 20.8 mL of 0.557 M H_2SO_4 solution? (20.2 mL)

10. Calculate the mass of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ required to prepare 500.0 mL of a 0.200M solution. (12.6 g)

11. Calculate the mass of $\text{Cu}_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ required to prepare 200.0 mL of a 0.300M solution. (19.9 g)

12. In a titration 0.200 M NaOH is used to neutralize 10.0 mL of H_2SO_4 . In three runs, the following data was collected. Calculate the concentration of the acid. (0.028 M)

	Volume of 0.20 M NaOH (mL)		
Initial Burette Reading	12.90	15.70	18.50
Final Burette Reading	15.70	18.50	21.50

13. Calculate the volume of 0.500M H_3PO_4 required to neutralize 25.0 mL of 0.200M NaOH. (0.00333 L)

(3.33 mL)

14. Calculate the volume of 0.50 M NaOH required to neutralize 35.0 mL of 0.100M $\text{H}_2\text{C}_2\text{O}_4$. (0.014 L)